

HW07_Lab_22125096

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```
setwd("D:/STAT/Lab/Lab7")  
getwd()
```

```
## [1] "D:/STAT/Lab/Lab7"
```

```
# Load the data  
data <- read.csv("manuf.csv",header=TRUE)  
# Log-transform the data  
log_Y <- log(data$y)  
log_K <- log(data$k)  
log_L <- log(data$l)  
log_E <- log(data$e)  
log_M <- log(data$m)
```

$$\text{Model } Y = \alpha K^{\beta_2} L^{\beta_3} E^{\beta_4} M^{\beta_5} \exp\{e\}$$

$$\Rightarrow \text{Model } Y = \ln \alpha + \beta_2 \ln K + \beta_3 \ln L + \beta_4 \ln E + \beta_5 \ln M + \epsilon$$

```
model <- lm(log_Y~log_K+log_L+log_E+log_M)
```

$$\widehat{SPRICE} = 11154.29 + 10680.00 \times \text{LIVAREA} - 11.33 \times \text{AGE} \\ - 15552.44 \times \text{BEDS} - 7019.30 \times \text{BATHS}$$

a) Hypothesis: